Summary in Graph

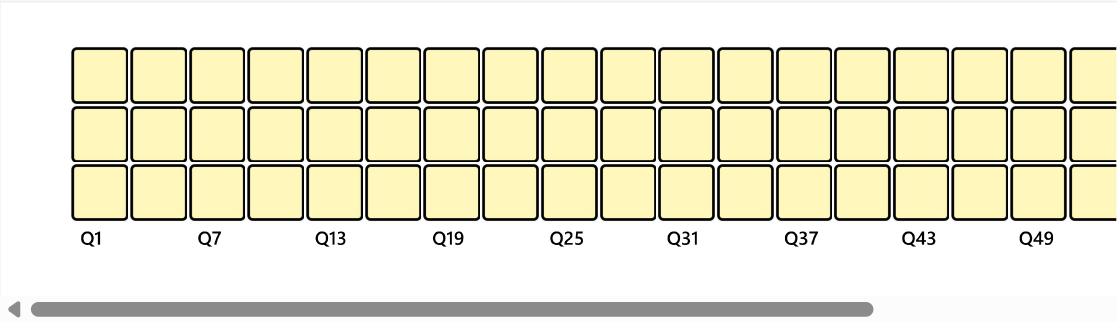
Exam Summary (GO Classes DA 2026 | All India Mock Test 5)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	65 (30 + 35)
Total Marks:	100 (30 + 70)
Exam Duration:	180 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



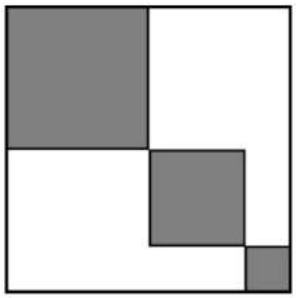
* Average Stats based on first-attempt stats of Top 10% (27) out of 270 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Aptitude

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Analytical Aptitude

In the diagram, a square with side length 6 is partially shaded.



The largest shaded region is a square with side length 3 . The other two shaded regions are squares with side lengths 2 and 1. What is the total area of the unshaded region?

A. 12

B. 18

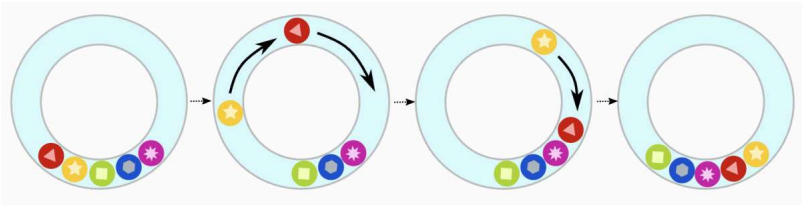
C. 22

D. 24

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

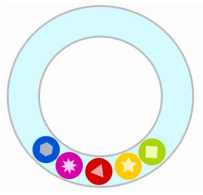
Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Spatial Aptitude

A baby's rattle is made of a clear circular tube with colourful balls inside. When the baby shakes the rattle, some balls can move through the tube. An example of the rattle before and after it is shaken is shown.

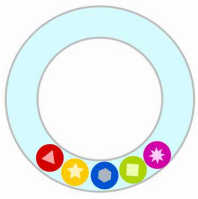


While there is room for the balls to move, there isn't room for one ball to move past another ball. Which of the following could be the baby's rattle after it has been shaken again?

A.



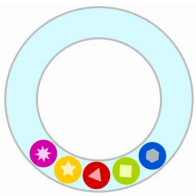
B.



C.



D.



Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

A container of ice cream can make 6 cones or it can make 4 sundaes. If 5 such containers of ice cream are used to make 12 cones, what is the greatest number of sundaes that can be made with the ice cream that remains?

A. 4

B. 8

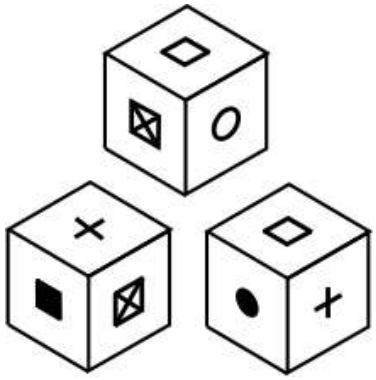
C. 12

D. 16

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Spatial Aptitude

Three different views of the same cube are shown.



The symbol on the face opposite ● is

A.



B.



C.



D.



Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

At the start of this month, Mathilde and Salah each had 100 coins. For Mathilde, this was 25% more coins than she had at the start of last month. For Salah, this was 20% fewer coins than he had at the start of last month. The total number of coins that they had at the start of last month was

A. 180

B. 185

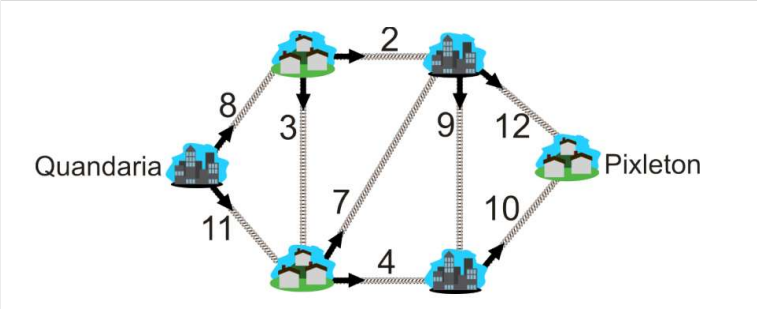
C. 190

D. 205

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Analytical Aptitude

In the land of Bebravia, there are six towns. All food is grown in **Quandaria** and transported by train to the other towns. The train routes are indicated by lines drawn between towns in the diagram. The arrow on each line indicates the direction in which trains carrying food are allowed to move.



Bebravia is not large, so there is more than enough time for many trains to travel between any two towns. The only limitation is the daily capacity on each train route: the number on each line indicates the maximum number of trains that can travel along that route in one day.

In one day, what is the **largest number of trains** that can travel from **Quandaria** to **Pixleton**?

A. 13

B. 16

C. 19

D. 12

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 08sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

There are 20 pens to be given away to 4 students. Each student receives a different number of pens and each student receives at least one pen. What is the largest number of pens that a student can receive?

A. 17

B. 15

C. 14

D. 8

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

At the beginning of the first day, a box contains 1 black ball, 1 gold ball, and no other balls. At the end of each day, for each gold ball in the box, 2 black balls and 1 gold ball are added to the box; this means that at the end of the first day, there are 5 balls in the box. If no balls are removed from the box, how many balls are

in the box at the end of the seventh day?

A. 395

B. 371

C. 389

D. 383

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

Amy bought and then sold $15n$ avocados, for some positive integer n. She made a profit of $\$100$. (Her profit is the difference between the total amount that she earned by selling the avocados and the total amount that she spent in buying the avocados.) She paid $\$2$ for every 3 avocados. She sold every 5 avocados for $\$4$. What is the value of n ?

A. 100

B. 20

C. 50

D. 30

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Verbal Aptitude

Choose the most appropriate alternative from the options given below to complete the following sentence:

If the tired soldier wanted to lie down, he _____ the mattress out on the balcony.

A. should take

B. shall take

C. should have taken

D. will have taken

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Technical

Q #1 | Multiple Select Type | Award: 1 | Penalty: 0 | Artificial Intelligence

Consider performing tree search for some search graph. Let depth (n) be the depth of search node n and $\text{cost}(n)$ be the total cost from the start state to node n. Let G_d be a goal node with minimum depth, and G_c be a goal node with minimum total cost.

For iterative deepening (where we repeatedly run DFS and increase the maximum depth allowed by 1), mark all conditions that are guaranteed to be true for every node n that could be expanded during the search.

A. $\text{cost}(n) \leq \text{cost}(G_c)$

B. $\text{cost}(n) \leq \text{cost}(G_d)$

C. $\text{depth}(n) \leq \text{depth}(G_c)$

D. $\text{depth}(n) \leq \text{depth}(G_d)$

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Artificial Intelligence

Consider heuristics where $h(n) = 0$ for all n such that $\text{GOAL}(n) = \text{True}$, i.e., heuristics that assign value zero to goal states.

The following statements are either true or false:

Statement 1 : Every admissible heuristic is consistent.
Statement 2 : Every consistent heuristic is admissible.

Using the standard definition of consistency, with n' a successor of state n and a an action, a consistent heuristic must satisfy $h(n) \leq c(n, a, n') + h(n')$.

Which of the following options is correct?

A. Statement 1 is true and Statement 2 is true.

B. Statement 1 is true and Statement 2 is false.

C. Statement 1 is false and Statement 2 is true.

D. Statement 1 is false and Statement 2 is false.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

Select all that apply: Given a set of input features x, where $x \in \mathbb{R}^n$, you are tasked with predicting a label for y, where $y = 1$ or $y = -1$. You have no knowledge of about the distribution of x and of y. Which of the following methods are appropriate?

A. Perceptron

B. k-Nearest Neighbors

C. Linear Regression

D. Decision Tree with unlimited depth

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Recall that in regularization, given an objective function $J(\Theta)$, the parameter estimate is obtained as

$$\hat{\Theta} = \arg\min_{\Theta} J(\Theta) + \lambda r(\Theta),$$

where $r(\Theta)$ is the regularization term. Which of the following statements is correct?

A. The recommended way to choose λ is to pick the value of λ that minimizes the training set error.

B. The recommended way to choose λ is to pick the value of λ that minimizes the validation error.

C. The recommended way to choose λ is to pick the value of λ that minimizes the test set error.

D. The recommended way to choose λ is to pick the largest possible value of λ to avoid overfitting.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider the function

$$f : \mathbb{R}^2 \to \mathbb{R}, \quad f(\mathbf{w}) = \log\left(1 + e^{-\mathbf{w}^T\mathbf{x}}\right),$$

where $\mathbf{w} = (w_1, w_2)^T \in \mathbb{R}^2$ and $\mathbf{x} = (x_1, x_2)^T \in \mathbb{R}^2$ is a constant vector.

What is the gradient $\nabla f(\mathbf{w})$?

A. $\nabla f(\mathbf{w}) = -\dfrac{1}{1 + e^{\mathbf{w}^T\mathbf{x}}}\,\mathbf{x}$

B. $\nabla f(\mathbf{w}) = \dfrac{1}{1 + e^{-\mathbf{w}^T\mathbf{x}}}\,\mathbf{x}$

C. $\nabla f(\mathbf{w}) = -\dfrac{e^{\mathbf{w}^T\mathbf{x}}}{1 + e^{\mathbf{w}^T\mathbf{x}}}\,\mathbf{x}$

D. $\nabla f(\mathbf{w}) = -\dfrac{1}{1 + e^{\mathbf{w}^T\mathbf{x}}}\,\mathbf{w}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider a dataset of points $x_1, \ldots, x_n$ with labels $y_1, \ldots, y_n \in \{-1, 1\}$, such that the data is separable. We run a soft-margin SVM and a hard-margin SVM, and in each case obtain parameters w and b. Pick **one** that is true:

A. The resulting w and b are the same in the two cases, hence the boundaries are the same.

B. The resulting w and b can be different in the two cases, but the boundaries are the same.

C. The resulting w and b can be different, and the boundaries can be different.

D. None of the above.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider a dataset whose average value is $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$.

The loss for a single data point y is defined as

$$L(y, h) = (\alpha y - h)^2 + \lambda h,$$

where α and λ are constants.

The average loss over the dataset is

$$L(h) = \frac{1}{n}\sum_{i=1}^{n} L(y_i, h).$$

Which of the following is the value of h^* that minimizes $L(h)$?

A. $h^* = \alpha\bar{y}$

B. $h^* = \alpha\bar{y} - \dfrac{\lambda}{2}$

C. $h^* = \alpha\bar{y} + \dfrac{\lambda}{2}$

D. $h^* = \dfrac{\alpha\bar{y} - \lambda}{2}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let X be a discrete random variable with probability mass function

$$P(X = x) = k(1 + |x|)^2,$$

for $x \in \{-2, -1, 0, 1, 2\}$, where k is a real constant chosen so that $P(X = x)$ is a valid probability mass function.

Then the value of $P(X = 0)$ is:

A. $1/9$

B. $2/27$

C. $1/27$

D. $1/81$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let $X \sim \text{Poisson}(\lambda)$ and suppose

$$P(X = 1) + 2P(X = 0) = 12P(X = 2).$$

Which of the following equations must be satisfied by λ?

A. $3\lambda^2 - \lambda - 2 = 0$
B. $6\lambda^2 - \lambda - 2 = 0$
C. $6\lambda^2 - 2\lambda - 1 = 0$
D. $12\lambda^2 - \lambda - 2 = 0$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let X and Y have the joint probability mass function

$$P(X = x, Y = y) = \frac{1}{3x}, \quad y = 1, 2, \ldots, x; \quad x = 1, 2, 3.$$

Then the value of the conditional expectation $E(Y \mid X = 3)$ is

A. 1

B. 2

C. 1.5

D. 2.5

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let $X_1, X_2, \ldots, X_n (n \geq 3)$ be a random sample from a $N(\mu, \sigma^2)$ distribution, where $\mu \in \mathbb{R}$ and $\sigma > 0$ are both unknown. Then which one of the following is a simple null hypothesis?

A. $H_0 : \mu < 5, \sigma^2 = 3$

B. $H_0 : \mu = 5, \sigma^2 > 3$

C. $H_0 : \mu = 5, \sigma^2 = 3$

D. $H_0 : \mu = 5$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let $Ax = b$ be a non-homogeneous system of linear equations. The augmented matrix $[A : b]$ is given by

$$\left[\begin{array}{cccc|c} 1 & 1 & -2 & 1 & 1 \\ -1 & 2 & 3 & -1 & 0 \\ 0 & 3 & 1 & 0 & -1 \end{array}\right]$$

Which of the following statements is true?

A. Rank of A is 3.
B. The system has no solution.
C. The system has a unique solution.
D. The system has infinitely many solutions.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

If A is a 3×3 non-zero matrix such that $A^2 = 0$, then the number of non-zero eigenvalues of A is

A. 0

B. 1

C. 2

D. 3

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let $P = (p_{ij})$ be a 50×50 matrix, where $p_{ij} = \min(i, j)$, $i, j = 1, 2, \ldots, 50$. Then the rank of P equals

A. 1
B. 2
C. 25
D. 50

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let A be a 3×3 real matrix with eigenvalues $1, 2, 3$ and let $B = A^{-1} + A^2$. Then the trace of the matrix B is equal to

A. $\frac{91}{6}$

B. $\frac{95}{6}$

C. $\frac{97}{6}$

D. $\frac{101}{6}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let $A = \begin{pmatrix} 1 & 1 & 1 \\ 3 & -1 & 1 \\ 1 & 5 & 3 \end{pmatrix}$ and V be the vector space of all $X \in \mathbb{R}^3$ such that $AX = 0$. Then $\dim(V)$ is

A. 0

B. 1

C. 2

D. 3

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Numerical Type | Award: 1 | Penalty: 0 | Calculus

For real constants a and b, let

$$f(x) = \begin{cases} \dfrac{a \sin x - 2x}{x}, & x < 0 \\ bx, & x \geq 0 \end{cases}$$

If f is a differentiable function, then the value of $a + b$ is

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider the following algorithm:

ALG1$(A[1 \ldots n])$:

- For $i = 1$ to $n - 1$:
  - For $j = 1$ to $i - 1$:
    - If $A[j] > A[i]$, swap the values of $A[j]$ and $A[i]$.

Which of the following statements is true at the beginning of iteration i of the outer loop?

A. A is sorted

B. $A[1 \ldots (i-1)]$ is sorted

C. $A[1 \ldots (i-1)]$ is sorted and $x \leq A[i]$ for all $x \in A[1 \ldots (i-1)]$

D. $A[1 \ldots (i-1)]$ is sorted and $x \leq y$ for all $x \in A[1 \ldots (i-1)]$ and $y \in A[i \ldots n]$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Suppose $f_1(n)$ is $O(n^2)$ and $\Omega(n)$. Also suppose that $f_2(n) = \Theta(n^2)$.

The following statements are either true or false:

Statement 1 : The function $f(n) = f_1(n) + f_2(n)$ must be $\Theta(n^2)$.

Statement 2 : The function $g(n) = \dfrac{f_2(n)}{f_1(n)}$ must be $\Omega(n)$.

Which of the following options is correct?

A. Statement 1 is true and Statement 2 is true.

B. Statement 1 is true and Statement 2 is false.

C. Statement 1 is false and Statement 2 is true.

D. Statement 1 is false and Statement 2 is false.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

A queue Q contains n elements, and an empty stack S is available.

You need to transfer all elements from Q to S in such a way that the front element of the queue becomes the top element of the stack, and the relative order of all elements is preserved.

You are allowed to perform only the following operations:

- On the queue: Enqueue, Dequeue
- On the stack: Push, Pop

What is the minimum number of Enqueue operations required to complete this transfer?

A. 0

B. $n - 1$

C. n

D. $2n$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Consider the partition operation used in QuickSort. A partition selects a pivot and rearranges the array so that all elements to the left of the pivot are ≤ pivot and all elements to the right are ≥ pivot.

Which of the following arrays could have been partitioned at least once? Select all that apply.

A. $[50, 10, 20, 30, 60, 40]$

B. $[20, 10, 30, 60, 40, 50]$

C. $[20, 50, 10, 40, 30, 60]$

D. $[60, 50, 40, 30, 20, 10]$

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in Python

Consider the following Python code:

```
lst = ['A', 'B', 'C']
lst[1] = ['X', 'Y', 'Z']
x = lst[1]
y = x
y[0] = 'P'
```

What is the final value of `lst`?

A. `['A', ['X','Y','Z'], 'C']`
B. `['A', ['P','Y','Z'], 'C']`
C. `['A', ['X','P','Z'], 'C']`
D. `Error`

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Consider a table People with attributes SALARY and NAME. Given the relational algebra expression $\Pi_{\text{NAME}}\ (\sigma_{\text{SALARY}>10{,}000}\ (\Pi_{\text{NAME}\cup\text{SALARY}}\ (\text{People}))$, which of the following are valid rewrites? (Check all that apply.)

A. $\Pi_{\text{NAME}}\ (\sigma_{\text{SALARY}>10{,}000}(\text{People}))$

B. $\Pi_{\text{NAME}}\ (\sigma_{\text{SALARY}>10{,}000}\ (\Pi_{\text{SALARY}}\ (\text{People}))$

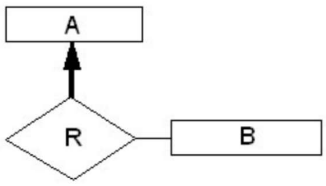
C. $\sigma_{\text{SALARY}>10{,}000}\ (\Pi_{\text{NAME U SALARY}}\ (\text{People})$

D. $\sigma_{\text{SALARY}>10{,}000}\ (\Pi_{\text{NAME}}\ (\Pi_{\text{NAME U SALARY}}\ (\text{People}))$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Select Type | Award: 1 | Penalty: 0 | Databases

Given the portion of an ER diagram shown below (with a thickened arrow from R to A), which of the following statements are true? (Select all that apply)



A. R connects each entity in A to at least one entity in B

B. R connects each entity in A to at most one entity in B

C. R connects each entity in B to at least one entity in A

D. R connects each entity in B to at most one entity in A

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Artificial Intelligence

We use the dictionary:

M	domain of men
s	Sharon
$N(x)$	x is nice
$L(x, y)$	x loves y

Which of the following formulas corresponds to the sentence:

There is a nice man who loves Sharon.

A. $\exists x \in M[N(x) \wedge L(x, s)]$

B. $\exists x \in M[N(x) \rightarrow L(x, s)]$

C. $\forall x \in M[N(x) \wedge L(x, s)]$

D. $\forall x \in M[N(x) \rightarrow L(x, s)]$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

You are given a rooted complete binary tree T with n vertices. Each vertex v has a distinct integer label x_v. A vertex is called a *local minimum* if its label is smaller than the labels of all its neighbors (neighbors include the parent and the children).

Consider the following algorithm:

Start at the root.
If the current vertex is a local minimum, stop.
Otherwise, move to a child with a smaller label and repeat.

Which of the following statements are correct?

A. The algorithm always terminates at a local minimum.

B. The running time of the algorithm is $O(\log n)$.

C. The algorithm may revisit a vertex more than once during execution.

D. Along the execution of the algorithm, the labels of the visited vertices form a strictly decreasing sequence.

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

The input is a sequence of n integers with many repetitions. The number of distinct integers is k.

You may assume that the frequency of each distinct value has already been computed and stored (for example, in a hash table of size k), and retrieving the frequency of any value takes $O(1)$ time.

Which of the following statements is correct?

A. Any comparison-based sorting algorithm used to sort such an array must take $\Omega(n \log n)$ comparisons in the worst case.

B. The sequence can be sorted using $O(n + k \log k)$ time complexity.

C. The sequence can be sorted using $O(k \log n)$ time complexity.

D. The sequence can be sorted using $O(\log n)$ comparisons since the distinct values are already separated.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let A and B be two events such that $0 < P(A) < 1$ and $0 < P(B) < 1$.

Which one of the following statements is **NOT** true?

A. If $P(A \mid B) > P(A)$, then $P(B \mid A) > P(B)$

B. If $P(A \cup B) = 1$, then A and B cannot be independent

C. If $P(A \mid B) > P(A)$, then $P(A^c \mid B) < P(A^c)$

D. If $P(A \mid B) > P(A)$, then $P(A^c \mid B^c) < P(A^c)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Numerical Type | Award: 2 | Penalty: 0 | Probability

Let $Z_1, Z_2, \ldots, Z_{128}$ be independent and identically distributed Bernoulli random variables such that

$$P(Z_i = 1) = 0.5 \quad \text{and} \quad P(Z_i = 0) = 0.5.$$

Define the column vectors

$$X = \begin{bmatrix} Z_1 \\ Z_2 \\ \vdots \\ Z_{64} \end{bmatrix}, \qquad Y = \begin{bmatrix} Z_{65} \\ Z_{66} \\ \vdots \\ Z_{128} \end{bmatrix}.$$

Find the value of $\mathbb{E}[X^T Y]$.

Your Answer: | Correct Answer: 16 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Recall that the *k*-means algorithm minimizes the following objective on a dataset $D = \{x_i\}_{i=1}^n$:

$$J(c, \mu) = \sum_{i=1}^n \|x_i - \mu_{c_i}\|_2^2$$

where c_i is scalar, and $c_i \in \{1, \ldots, k\}$ is the cluster assignment of point x_i, and μ_j is the centroid of cluster j.

Consider the extreme case where the number of clusters equals the number of data points, i.e., $k = n$. Which of the following statements are ALWAYS correct?

A. The optimal cluster assignment satisfies $c_i^* = i$ for all $i \in \{1, \ldots, n\}$.

B. The optimal centroid corresponding to cluster i is given by $\mu_i^* = x_i$.

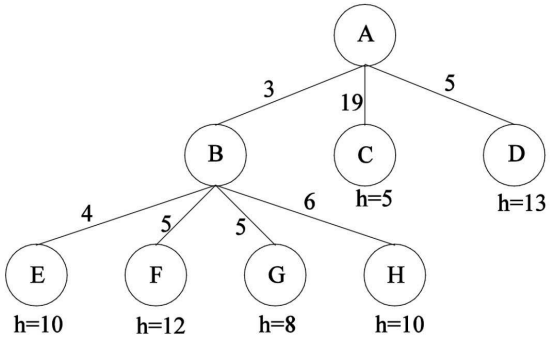
C. The optimal value of the objective function is $J(c^*, \mu^*) = 0$.

D. The optimal centroid is given by $\mu_i^* = \frac{1}{n} \sum_{j=1}^{n} x_j$.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #31 | Multiple Select Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Consider the following search tree produced after expanding nodes A and B. The tree structure is shown below.



Each arc is labeled with the cost of the corresponding operator, and each leaf node is labeled with the value of a heuristic function h.

For uninformed searches, assume:

- Children are expanded from left to right.
- In case of ties, nodes are expanded in alphabetical order.

Which of the following statements are correct regarding the next node to be expanded by the given search strategies?

A. Using Depth-First Search, the next node expanded is E.

B. Using Greedy Best-First Search, the next node expanded is C.

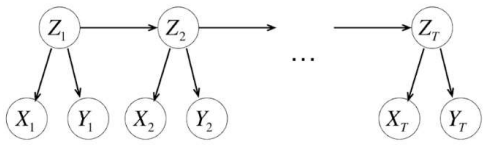
C. Using Uniform-Cost Search, the next node expanded is D.

D. Using A^* search, the next node expanded is G.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #32 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

Consider a variant of a hidden Markov model in which each hidden state Z_t generates two conditionally independent observations X_t, Y_t :



Suppose each hidden state Z_t takes one of K possible values, each observation X_t takes one of M_1 possible values, and each observation Y_t takes one of M_2 possible values. Assume the model is homogeneous, so that transition and emission probabilities are the same for all t. What is the total number of parameters in this model? Choose the tightest expression below.

A. $O\left(K^2 M_1 M_2\right)$

B. $O\left(K^2\left(M_1+M_2\right)\right)$

C. $O\left(K^2+K M_1 M_2\right)$

D. $O\left(K^2+K\left(M_1+M_2\right)\right)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #33 | Numerical Type | Award: 2 | Penalty: 0 | Machine Learning

Adam proposed a new algorithm called perceptron trees, which combines ideas from both decision trees and perceptrons. Perceptron trees work like decision trees, but instead of using a majority vote at each leaf node, a perceptron is used to make the final decision.

Consider a binary classification problem with six binary features $A, B, C, D, E, F \in \{0,1\}$. The internal nodes of the tree split on feature values (using features A, B, and D), and each leaf node contains a perceptron. The perceptron at a leaf uses the indicated subset of features (such as C, D, E, F or B, C, E, F) with the corresponding weights shown.

The structure of the perceptron tree and the weights of the perceptrons at the leaf nodes are shown in the figure below.

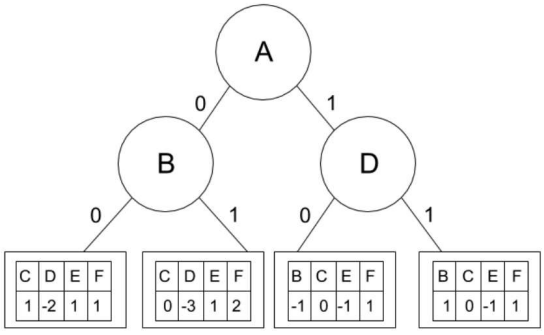


Figure 1: Perceptron Tree of max depth=2

Assume that the bias term for each perceptron is $+1$.

Given the input sample $[A, B, C, D, E, F] = [1, 1, 0, 1, 0, 1]$, predict the output label produced by the perceptron tree.

Enter 1 if the predicted label is $+1$, and enter -1 if the predicted label is -1.

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #34 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Consider the set of points $\{2^0, 2^1, 2^2, \ldots, 2^n\}$, where each point lies on the real number line at its corresponding numerical value. Hierarchical clustering is performed on these points using single-link (SL), complete-link (CL), and average-link (AL) methods.

Two distance functions are used:

1. Euclidean distance $d_E(A, B) = \|A - B\|_2$
2. Ratio-based distance $d_R(A, B) = \frac{\max(A,B)}{\min(A,B)}$

Let T_{SL}, T_{CL}, and T_{AL} denote the hierarchical clustering trees obtained using single-link, complete-link, and average-link methods respectively, under a given distance function.

Which of the following statements are correct?

A. Under Euclidean distance: $T_{SL} = T_{CL} = T_{AL}$

B. Under Euclidean distance: $T_{SL} = T_{CL} \neq T_{AL}$

C. Under ratio-based distance: $T_{SL} \neq T_{CL} = T_{AL}$

D. Under ratio-based distance: $T_{SL} \neq T_{CL} \neq T_{AL}$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #35 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider the objective function

$$f(w) = L(w) + \frac{\lambda}{2}\|w\|_2^2,$$

where $L(w)$ denotes the loss function.

We apply stochastic gradient descent (SGD) with batch size one and learning rate

$$\eta_t = \frac{1}{\lambda \cdot t}, \qquad t = 1, 2, 3, \ldots$$

Note that the learning rate keeps on changing at every iteration t.

Assume that the initial parameter vector is $w^{(1)} = 0$.

At iteration t, let g_t denote the gradient of the loss computed using the current sample.

Which of the following gives the expression for $w^{(t+1)}$ obtained after SGD updates?

A. $w^{(t+1)} = w^{(t)} - \eta_t \, g_t$

B. $w^{(t+1)} = w^{(t)} - \eta_t \left(g_t - \lambda w^{(t)}\right)$

C. $w^{(t+1)} = -\eta_t \sum_{j=1}^{t} g_j$

D. $w^{(t+1)} = -\frac{1}{t} \sum_{j=1}^{t} g_j$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #36 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

In PCA, suppose we sample $s \sim \mathcal{N}(0, \Sigma)$ and generate a new data point as $x = Ws$. What is the distribution of x?

A. $\mathcal{N}\left(0,\ W\Sigma W^\top\right)$
B. $\mathcal{N}(0,\ \Sigma)$
C. $\mathcal{N}\left(0,\ W^\top \Sigma W\right)$
D. $\mathcal{N}\left(0,\ WW^\top \Sigma\right)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #37 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider a binary classification problem where the test set has $N_{\text{pos}} > 0$ positive observations and $N_{\text{neg}} > 0$ negative observations. Suppose that we have previously trained a model on the training set, and that, on the test set, this model has a true positive rate value denoted by TPR and a false positive rate value denoted by FPR.

Which of the following is a correct expression for the accuracy of this model on the test set?

A. $\dfrac{N_{\text{pos}} \cdot \text{TPR} + N_{\text{neg}}(1 - \text{FPR})}{N_{\text{pos}} + N_{\text{neg}}}$

B. $\dfrac{N_{\text{pos}} \cdot (1 - \text{TPR}) + N_{\text{neg}}(1 - \text{FPR})}{N_{\text{pos}} + N_{\text{neg}}}$

C. $\dfrac{N_{\text{pos}} \cdot \text{TPR} + N_{\text{neg}} \cdot \text{FPR}}{N_{\text{pos}} + N_{\text{neg}}}$

D. $\dfrac{\text{TPR} + (1 - \text{FPR})}{2}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #38 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let $X_1, X_2, \ldots$ be i.i.d. Normal random variables with mean 2 and variance 3. Let N be a Poisson random variable with mean 4, independent of $\{X_1, X_2, \ldots\}$. Define

$$Y = X_1 + \cdots + X_N \quad \text{if } N \geq 1, \qquad Y = 0 \text{ if } N = 0.$$

What is the expected value $\mathbb{E}[Y]$?

A. 4
B. 8
C. 12

D. 16

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #39 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let X be a random variable uniformly distributed on $(0, 1)$. Suppose that, given X, the random variable Y is uniformly distributed on $(0, X)$. Given X and Y, the random variable Z is uniformly distributed on $(Y, 1)$.

What is the value of

$$\mathbb{E}[Z]\,?$$

A. $1/8$
B. $3/8$
C. $5/8$
D. $1/2$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #40 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let (X, Y) have the joint probability density function

$$f(x,y) = \begin{cases} \frac{1}{2}y^2 e^{-x}, & 0 < y < x < \infty \\ 0, & \text{otherwise} \end{cases}$$

Then $P(Y < 1 \mid X = 3)$ equals

A. $1/81$
B. $1/27$
C. $1/9$
D. $1/3$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #41 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let X be a random variable with the cumulative distribution function

$$F(x) = \begin{cases} 0, & x < 0 \\ x/8, & 0 \le x < 2 \\ x^2/16, & 2 \le x < 4 \\ 1, & x \ge 4 \end{cases}$$

Then $E(X)$ equals

A. 12/31
B. 13/12
C. 31/21
D. 31/12

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #42 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

An institute purchases laptops from either vendor V_1 or vendor V_2 with equal probability. The lifetimes (in years) of laptops supplied by vendor V_1 follow a $U(0,4)$ distribution, whereas the lifetimes (in years) of laptops supplied by vendor V_2 follow an $\mathrm{Exp}(1/2)$ distribution.

If a randomly selected laptop in the institute has a lifetime of more than two years, then the probability that it was supplied by vendor V_2 is

A. $\dfrac{2}{2+e}$
B. $\dfrac{1}{1+e}$
C. $\dfrac{1}{1+e^{-1}}$
D. $\dfrac{2}{2+e^{-1}}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #43 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be the linear transformation defined by $T(x, y, z) = (x + y, y + z, z + x)$ for all $(x, y, z) \in \mathbb{R}^3$. Then

A. $\mathrm{rank}(T) = 0, \mathrm{nullity}(T) = 3$

B. $\mathrm{rank}(T) = 2, \mathrm{nullity}(T) = 1$

C. $\mathrm{rank}(T) = 1, \mathrm{nullity}(T) = 2$

D. $\mathrm{rank}(T) = 3, \mathrm{nullity}(T) = 0$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #44 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Consider the subspace $W = \left\{(x_1, x_2, \ldots, x_{10}) \in \mathbb{R}^{10} : x_n = x_{n-1} + x_{n-2} \text{ for } 3 \leq n \leq 10\right\}$ of the vector space $\mathbb{R}^{10}$. The dimension of W is

A. 2

B. 3

C. 9

D. 10

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #45 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let $M = \begin{bmatrix} 0 & 1 & -2 \\ -1 & 0 & \alpha \\ 2 & -\alpha & 0 \end{bmatrix}$, $\alpha \in \mathbb{R} \backslash \{0\}$ and $\boldsymbol{b}$ a non-zero vector such that $M\boldsymbol{x} = \boldsymbol{b}$ for some $\boldsymbol{x} \in \mathbb{R}^3$. Then the value of $\boldsymbol{x}^T\boldsymbol{b}$ is

A. $-\alpha$

B. α

C. 0

D. 1

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #46 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let $M = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} \\ 0 & 1 \end{bmatrix}$ and $x = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. Then

$$\lim_{n \to \infty} M^n x$$

A. does not exist

B. is $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

C. is $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$

D. is $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #47 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let M be a 4×3 real matrix and let $\{e_1, e_2, e_3\}$ be the standard basis of $\mathbb{R}^3$. Which of the following is true?

A. If $\text{rank}(M) = 1$, then $\{Me_1, Me_2\}$ is a linearly independent set

B. If $\text{rank}(M) = 2$, then $\{Me_1, Me_2\}$ is a linearly independent set

C. If $\text{rank}(M) = 2$, then $\{Me_1, Me_3\}$ is a linearly independent set

D. If $\text{rank}(M) = 3$, then $\{Me_1, Me_3\}$ is a linearly independent set

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #48 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Let $f : [0, 4] \to \mathbb{R}$ be a twice differentiable function. Further, let $f(0) = 1, f(2) = 2$ and $f(4) = 3$. Then

A. there does NOT exist any $x_1 \in (0, 2)$ such that $f'(x_1) = \frac{1}{2}$

B. there exist $x_2 \in (0, 2)$ and $x_3 \in (2, 4)$ such that $f'(x_2) = f'(x_3)$

C. $f''(x) > 0$ for all $x \in (0, 4)$

D. $f''(x) < 0$ for all $x \in (0, 4)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #49 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Let f be a differentiable function defined on the interval $[0, 1]$. If $c \in (0, 1)$ is such that

$$f(x) < f(c) = f(0) \text{ for all } x \in (0, 1],\ x \neq c,$$

then which of the following is correct?

A. $f'(c) = 0$ and $f'(0) = 0$
B. $f'(c) = 0$ and $f'(0) \leq 0$
C. $f'(c) > 0$ and $f'(0) \leq 0$
D. $f'(c) = 0$ and $f'(0) > 0$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #50 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Partition the vertex set of a graph $G = (V, E)$ (with no self-loops) into two parts $V = V_1 \cup V_2$ randomly as follows:

- Each vertex $v \in V$ flips a coin independently.
- If the outcome is head, v joins V_1.
- If the outcome is tail, v joins V_2.

Suppose the coin is biased such that the probability of head is $2/3$ and the probability of tail is $1/3$.

What is the expected number of edges crossing between V_1 and V_2?

A. $3/9\,|E|$
B. $4/9\,|E|$
C. $5/9\,|E|$
D. $6/9\,|E|$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #51 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider a directed graph $G = (V, E)$ that is *unipathic*, i.e., for any two vertices $u, v \in V$, there is at most one simple path from u to v. Edge weights may be positive or negative. We wish to compute shortest-path distances from a source vertex s to all other vertices, and report if a negative-weight cycle is reachable from s.

Which of the following statements is **correct**?

A. A standard Bellman–Ford algorithm is required since BFS or DFS cannot detect negative-weight cycles.
B. Since the graph is unipathic, DFS or BFS can be used to compute shortest paths, and a negative-weight cycle can be detected if a relaxation improves an already visited vertex.
C. Dijkstra's algorithm works correctly because the graph has at most one simple path between any two vertices.
D. Negative-weight cycles cannot exist in a unipathic graph.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #52 | Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

We maintain a dynamic set D of integers using a binary search tree. Inserting an element into D takes constant time. Each node x stores an additional field

$$\text{size}(x) = \text{size}(\text{left}(x)) + \text{size}(\text{right}(x)) + 1,$$

where NIL nodes have size 0.

The operation COUNT(D, x) returns the number of elements in D that are **strictly larger than** x. Starting from the root, the algorithm compares x with the current node key and moves left or right, updating a counter using the stored size values.

Which of the following statements are correct?

A. When $x < \text{key}(r)$ at a node r, adding $\text{size}(\text{right}(r)) + 1$ to the counter correctly counts elements larger than x.
B. The COUNT operation modifies the structure of the tree.
C. The running time of COUNT is proportional to the height of the tree.
D. COUNT cannot be executed correctly if x is not present in D.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #53 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in Python

Consider the following Python code:

```
def f(n, d):
  if n in d:
    return d[n]
  if n <= 1:
    d[n] = n
  else:
    d[n] = f(n-1, d) + f(n-2, d)
  return d[n]
```

Suppose the following statements are executed:

```
D = {}
print(f(4, D))
print(D)
```

What is the output?

A. `3 and {4:3}`
B. `3 and {0:0, 1:1, 2:1, 3:2, 4:3}`
C. `4 and {0:0, 1:1, 2:2, 3:3, 4:4}`
D. `Error`

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #54 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Assume there are M records for $\mathrm{B}+$ tree to index. In the first case, there are N separate exact search queries on this index. In the second case, there is a range query on this index with results spanning N leaf nodes. How many pointers in the $\mathrm{B}+$ tree will these two cases traverse separately to retrieve the results? The first case: ____ ? ____ , the second case: ____ ? ____ .

A. $\Theta(\mathrm{N}+\log \mathrm{M}), \Theta(\mathrm{M}+\log \mathrm{N})$

B. $\Theta(\mathrm{N} \log \mathrm{M}), \Theta(\mathrm{M}+\log \mathrm{N})$

C. $\Theta(\mathrm{MN}), \Theta(\mathrm{N}+\log \mathrm{M})$

D. $\Theta(\mathrm{N} \log \mathrm{M}), \Theta(\mathrm{N}+\log \mathrm{M})$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #55 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Consider the attributes $\mathrm{A}\ \mathrm{B}\ \mathrm{C}\ \mathrm{D}\ \mathrm{E}\ \mathrm{F}$ which have the following functional dependencies:

$\mathrm{AD} \rightarrow \mathrm{F}$

$\mathrm{AE} \rightarrow \mathrm{G}$

$\mathrm{DF} \rightarrow \mathrm{BC}$

$\mathrm{E} \rightarrow \mathrm{C}$

$\mathrm{G} \rightarrow \mathrm{E}$

Consider the decomposition into 3 relations: $(\mathrm{ADF})(\mathrm{EC})(\mathrm{ABDEG})$.

This decomposition is ? (Select all that apply)

A. in BCNF

B. in $3\mathrm{NF}$

C. Dependency Preserving

D. Lossless

Your Answer: | **Correct Answer: D** | Not Attempted | Time taken: 00min 00sec | Discuss

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